

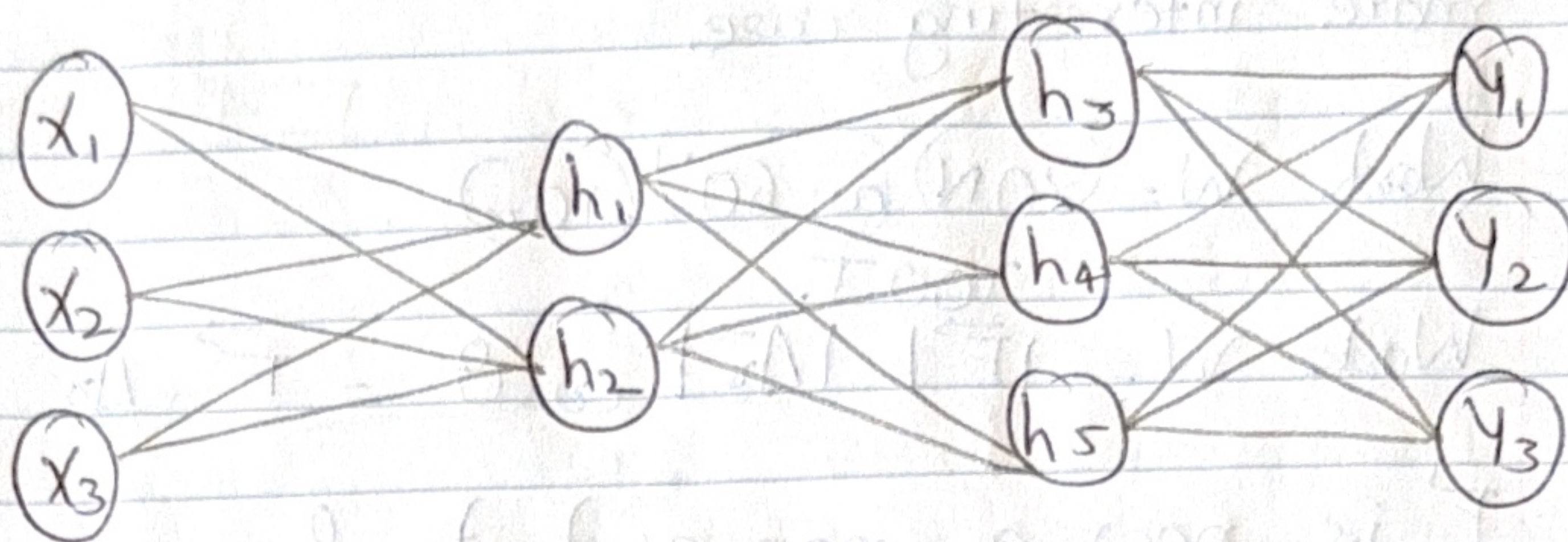
Home Work-3

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Problem 1)

Input layer : 3 nodes
 Hidden layer 1 : 2 nodes (ReLU)
 Hidden layer 2 : 3 nodes (ReLU)
 Output layer : 3 nodes (identity)

Input layer (3 nodes) Hidden layer 1 (2 nodes, ReLU) Hidden layer 2 (3 nodes, ReLU) Output layer (3 nodes, identity)



$$\rightarrow z^{(1)} = W^{(1)}x + b^{(1)} \quad \text{layer 1}$$

$$h^{(1)} = \text{ReLU}(z^{(1)})$$

$$z^{(2)} = W^{(2)}h^{(1)} + b^{(2)} \quad \text{layer 2}$$

$$h^{(2)} = \text{ReLU}(z^{(2)})$$

$$f(x; \theta) = W^{(3)}h^{(2)} + b^{(3)} \quad \text{Output}$$

Dimensions of all weight & bias matrices.

Matrix	Dimensions	Why
W^1	2×3	Maps 3 input $\rightarrow$ 2 hidden node
b^1	2×1	One bias per hidden node in layer 1
W^2	3×2	Maps 2 hidden $\rightarrow$ 3 hidden nodes
b^2	3×1	One bias per hidden node in layer 2
W^3	3×3	Maps 3 hidden $\rightarrow$ 3 output nodes.
b^3	3×1	One bias per output node.

$\rightarrow$ Total number of parameters:

Component	Calculation	Count.
W^1	2×3	6
b^1	2	2
W^2	3×2	6
b^2	3	3
W^3	3×3	9
b^3	3	3
Total		29

Eg) let $x = [1, 0, -1]$ & Weight = 0.5 & bias = 0

$$\rightarrow z' = w'x = \begin{bmatrix} 0.5 & 0.5 & 0.5 \\ 0.5 & 0.5 & 0.5 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

$$= [0.5, 0.5]$$

$$h' = \text{ReLU}([0.5, 0.5]) = [0.5, 0.5]$$

$\Rightarrow$ Both (+ve), unchanged.

$$\rightarrow z^2 = w^2 h' = \begin{bmatrix} [0.5, 0.5], \\ [0.5, 0.5], \\ [0.5, 0.5] \end{bmatrix} \times \begin{bmatrix} [0.5] \\ [0.5] \\ [0.5] \end{bmatrix} = \begin{bmatrix} [0.5] \\ [0.5] \\ [0.5] \end{bmatrix}$$

$$h^2 = \text{ReLU}([0.5, 0.5, 0.5]) = [0.5, 0.5, 0.5]$$

$$\rightarrow f(x; \theta) = w^3 h^2 = \begin{bmatrix} [0.5, 0.5, 0.5] \\ [0.5, 0.5, 0.5] \\ [0.5, 0.5, 0.5] \end{bmatrix} \times \begin{bmatrix} [0.5] \\ [0.5] \\ [0.5] \end{bmatrix}$$

$$= \begin{bmatrix} 0.75 \\ 0.75 \\ 0.75 \end{bmatrix}$$

$$\text{Final} = [0.75, 0.75, 0.75]$$

Problem-2) Since all observations are independent

$$\prod_{i=1}^n P(Y = y_i | X = x_i) = \prod_{i=1}^n \frac{e^{-f_0(x_i)} \cdot f_0(x_i)^{y_i}}{y_i!}$$

Using log rules: $\log(A \cdot B / C) = \log(A) + \log(B) - \log(C)$:

$$= \left[\sum_{i=1}^n \left[\log(e^{-f_0(x_i)}) + \log(f_0(x_i)^{y_i}) - \log(y_i!) \right] \right]$$

$$= \sum_{i=1}^n \left[-f_0(x_i) + y_i \log f_0(x_i) - \log(y_i!) \right]$$

→ The term $\log(y_i!)$ doesn't depend on θ at all. It is fixed by the data, so maximizing the log likelihood is equivalent to minimizing its negative, ignoring the constant: →

$$\hat{\theta} = \arg \min_{\theta} \sum_{i=1}^n \left[-f_{\theta}(x_i) + y_i \log f_{\theta}(x_i) \right] \quad (-1)$$

$$= \arg \min_{\theta} \sum_{i=1}^n \left[f_{\theta}(x_i) - y_i \log f_{\theta}(x_i) \right]$$

For ex) We have $n=2$ observations

i	y_i (actual count)	$f_{\theta}(x_i)$ (Model Prediction)
1	3	2.5
2	1	1.2

→ For $i = 1$

$$-f(x_1) + Y_1 \cdot \log f(x_1) - \log(Y_1!)$$

$$= (-2.5) + (3 \cdot \log(2.5)) - \log(3!)$$

$$= (-2.5) + (3(0.916)) - \log(6)$$

$$= (-2.5) + (2.748) - (1.792)$$

$$= -1.544$$

→ For $i = 2$

$$-f(x_2) + Y_2 \cdot \log f(x_2) - \log(Y_2!)$$

$$= (-1.2) + (1 \cdot \log(1.2)) - \log(1!)$$

$$= (-1.2) + 0.182 - 0$$

$$= -1.018$$

$$\text{Total log-likelihood} = (-1.544) + (-1.018)$$

$$= -2.562$$